Hotel Booking Demand Analytics Platform

Exploratory Data Analysis (EDA) Summary Report

Exploratory Data Analysis (EDA) was performed on the Hotel Booking Demand dataset to understand booking patterns, customer behavior, cancellation trends, revenue performance, and hotel operations. The analysis utilizes descriptive statistics and data visualization techniques to identify meaningful trends, relationships, and business insights. This report presents a comprehensive summary of the dataset, key business performance indicators, booking trends, customer characteristics, cancellation behavior, revenue analysis, stay patterns, and correlation analysis. The findings support data-driven decision-making and provide valuable recommendations for improving hotel operations and customer satisfaction.

1. Dataset Overview

The Hotel Booking Demand dataset contains hotel reservation records collected from City Hotels and Resort Hotels. It provides comprehensive information on customer bookings, hotel characteristics, reservation details, stay duration, cancellation status, pricing, and customer demographics. This dataset is widely used for hospitality analytics and supports the identification of booking patterns, customer behavior, and business performance.

The dataset consists of 119,390 booking records and 32 features, including numerical, categorical, and date-related variables. Before performing the exploratory data analysis, the dataset was thoroughly preprocessed by handling missing values, removing duplicate records, correcting data types, treating outliers, and engineering new features. The cleaned dataset was then used to generate meaningful visualizations and business insights.

Dataset Summary

Attribute	Details
Dataset Name	Hotel Booking Demand
Source	Kaggle
Total Records	119,390

Attribute	Details
Total Features	32
Hotel Types	City Hotel, Resort Hotel
Data Types	Numerical, Categorical, Date-Time
Target Variable	is_canceled
Data Quality	Cleaned, transformed, and validated for analysis

2. Business KPI Summary

The Business KPI Summary presents the most important performance indicators extracted from the Hotel Booking Demand dataset. These metrics provide a quick overview of booking volume, hotel performance, customer behavior, cancellation trends, pricing, and stay characteristics before the detailed exploratory data analysis.

Business KPI Summary

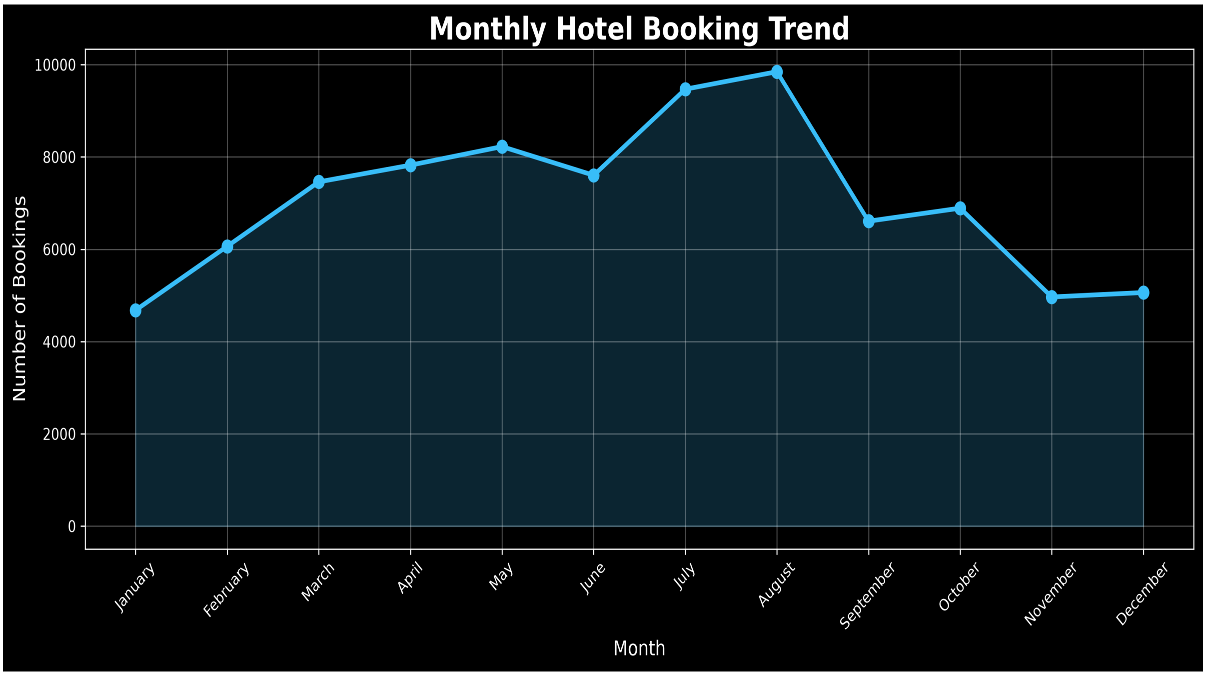
Business KPI	Value
Total Bookings	119,390
Total Customers	119,390
City Hotel Bookings	79,330
Resort Hotel Bookings	40,060
Cancellation Rate	37.04%
Average Daily Rate (ADR)	101.83
Average Stay Duration	3.43 Days
Average Lead Time	104.01 Days
Repeat Guest Percentage	3.19%
Average Special Requests	0.57

The above KPIs provide a concise summary of the overall hotel booking performance. They highlight booking volume, hotel distribution, cancellation behavior, pricing trends, customer loyalty, and stay characteristics, forming the foundation for the detailed exploratory presented in the following sections.

3. Booking Analysis

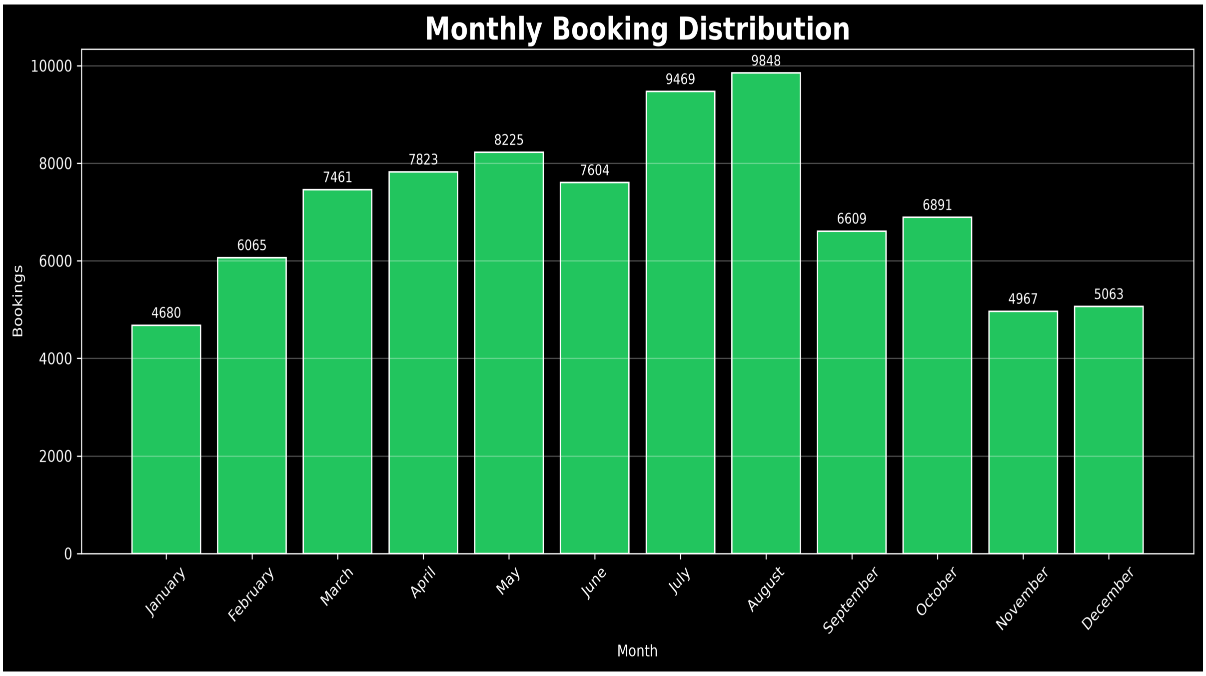
Booking analysis examines reservation patterns across different months, hotel types, booking lead times, and seasons. This analysis helps identify demand fluctuations, customer booking behavior, and peak booking periods that support operational planning and revenue optimization.

3.1 Monthly Booking Trend



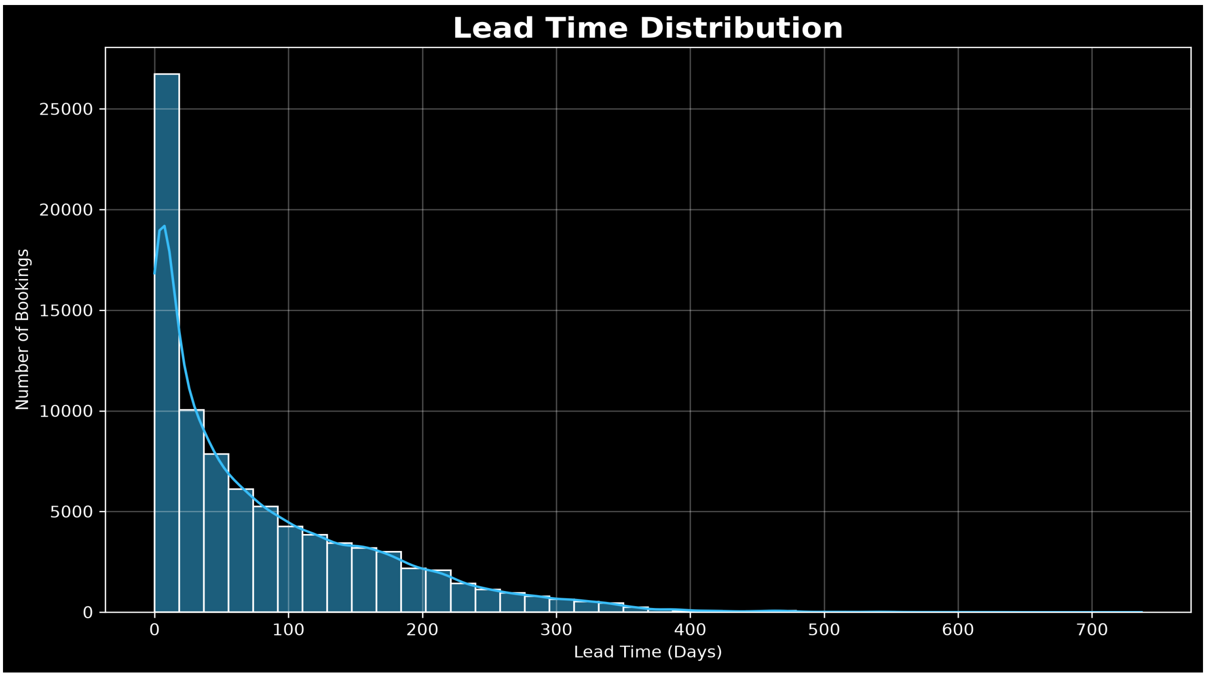
Monthly booking analysis shows seasonal variations in hotel demand. Booking activity increases during holiday seasons and decreases during off-peak months, helping hotels optimize staffing, inventory, and pricing strategies.

3.2 Hotel Type Comparison



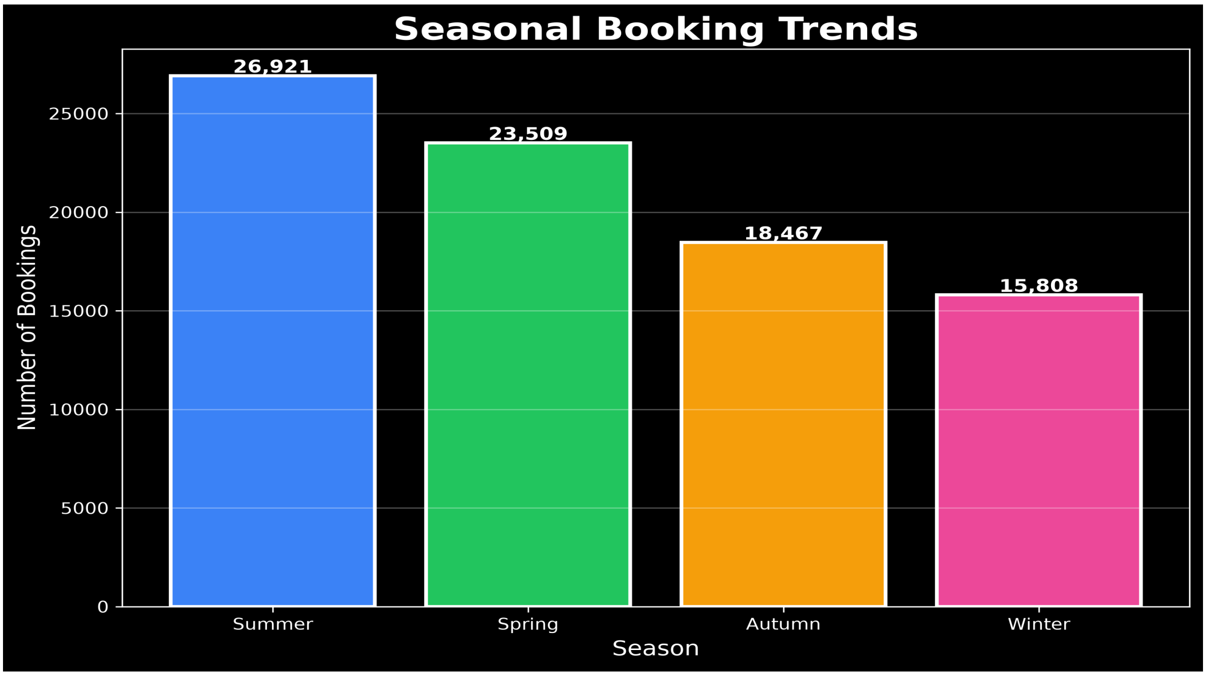
The hotel type comparison indicates that **City Hotels** account for **79,330** bookings, representing **61.97%** of all reservations. In comparison, **Resort Hotels** recorded **40,060** bookings, contributing **38.03%** of the total bookings. The results show that City Hotels receive a significantly higher proportion of reservations than Resort Hotels, suggesting stronger demand for urban accommodations due to business travel, accessibility, and city-based tourism. Resort Hotels continue to attract a substantial share of leisure travelers, particularly during vacation and holiday seasons.

3.3 Booking Lead Time Analysis



The booking lead time analysis shows that guests reserve their stays an average of **80.24** days before arrival, with booking periods ranging from **0** to **737** days. The wide variation in lead time indicates diverse booking behaviors, providing valuable insights for demand forecasting, pricing strategies, and inventory planning.

3.4 Seasonal Booking Trends

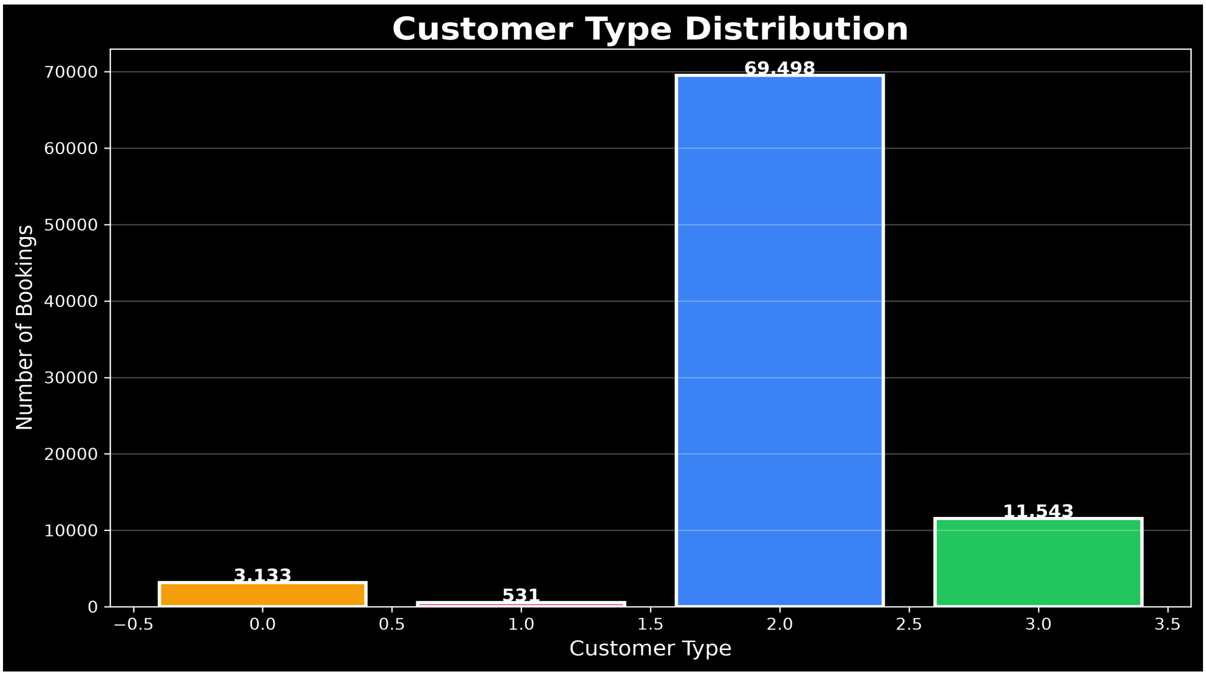


Seasonal booking analysis indicates that **Summer** recorded the highest number of bookings with **26,921** reservations, followed by **Spring** with **23,509**. In contrast, **Autumn** and **Winter** registered **18,467** and **15,808** bookings, respectively. These trends highlight seasonal fluctuations in hotel demand and provide valuable insights for capacity planning, promotional campaigns, and dynamic pricing strategies.

4. Customer Analysis

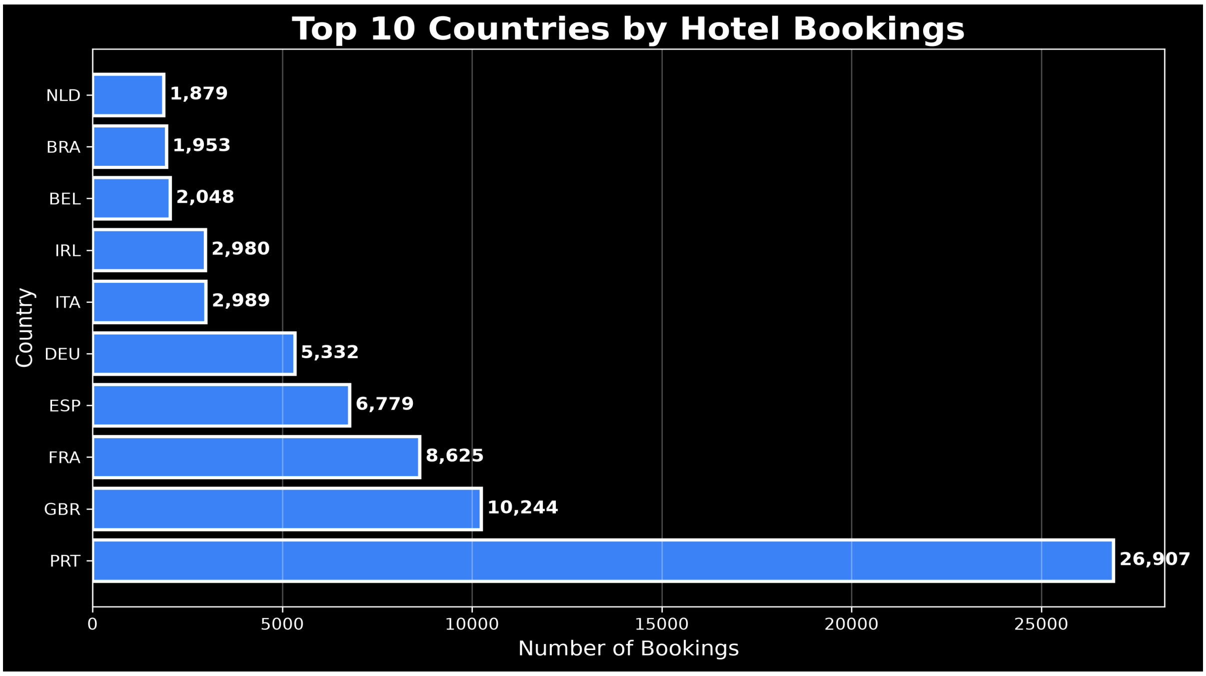
Customer Analysis explores guest characteristics and booking behavior to understand customer preferences and market demand. This section examines customer types, geographical distribution, repeat guest behavior, and market segments. The insights help hotels improve customer satisfaction, strengthen loyalty, and develop targeted marketing strategies.

4.1 Customer Type Distribution



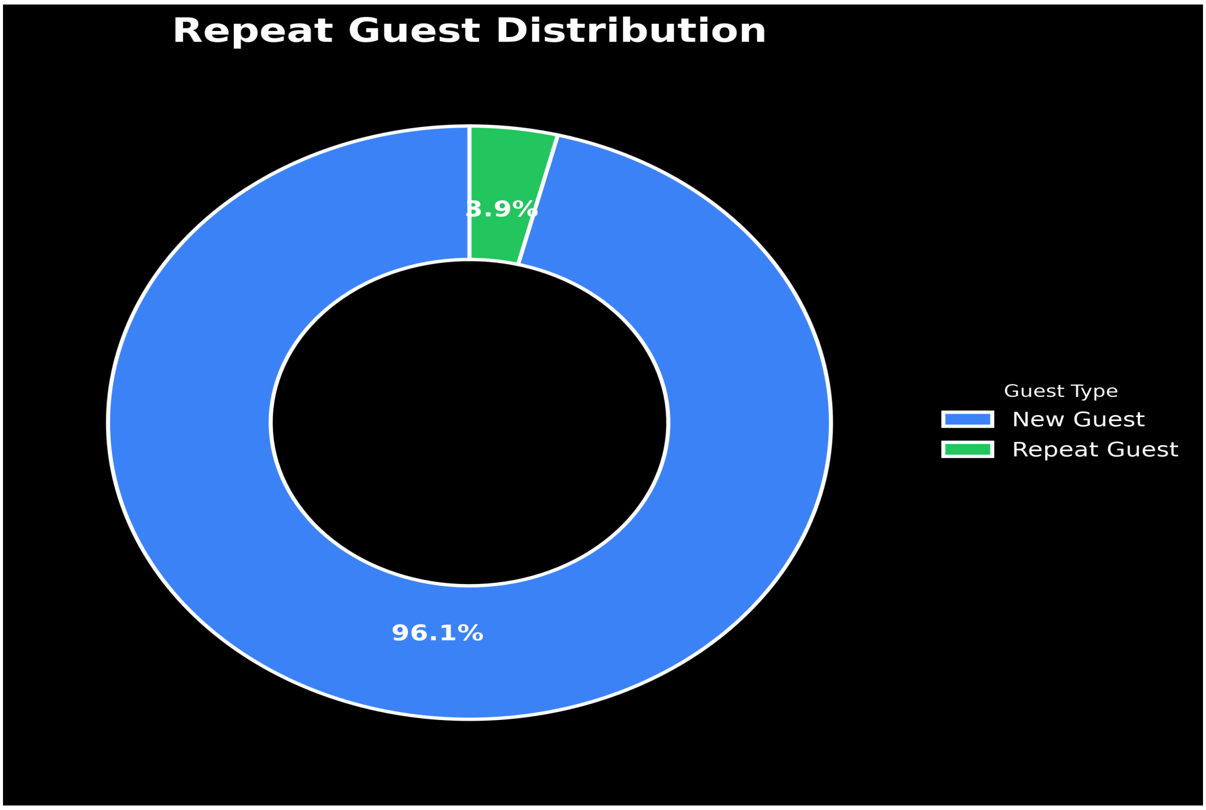
The majority of bookings were made by **Transient** customers (69,498), followed by **Transient-Party**, **Contract**, and **Group** customers. This indicates that individual travelers contribute the largest share of hotel bookings, highlighting the importance of focusing on independent guests.

4.2 Country-wise Bookings



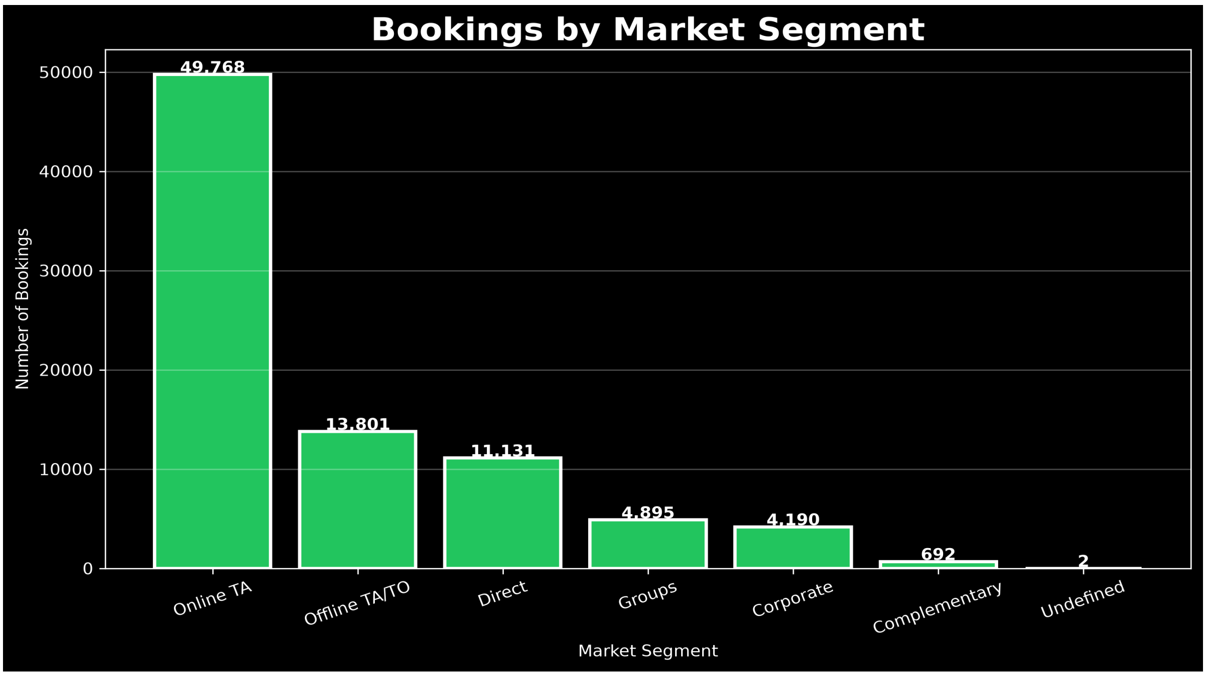
Portugal recorded the highest number of bookings with **26,907** reservations, followed by the United Kingdom (**10,244**), France (**8,625**), and Spain (**6,779**). The results indicate that hotel demand is primarily driven by domestic and European travelers.

4.3 Repeat Guest Analysis



The dataset contains **81,379** new guests and **3,326** repeat guests. The significantly higher proportion of new customers suggests an opportunity to strengthen customer loyalty through retention programs and personalized guest experiences.

4.4 Market Segment Analysis

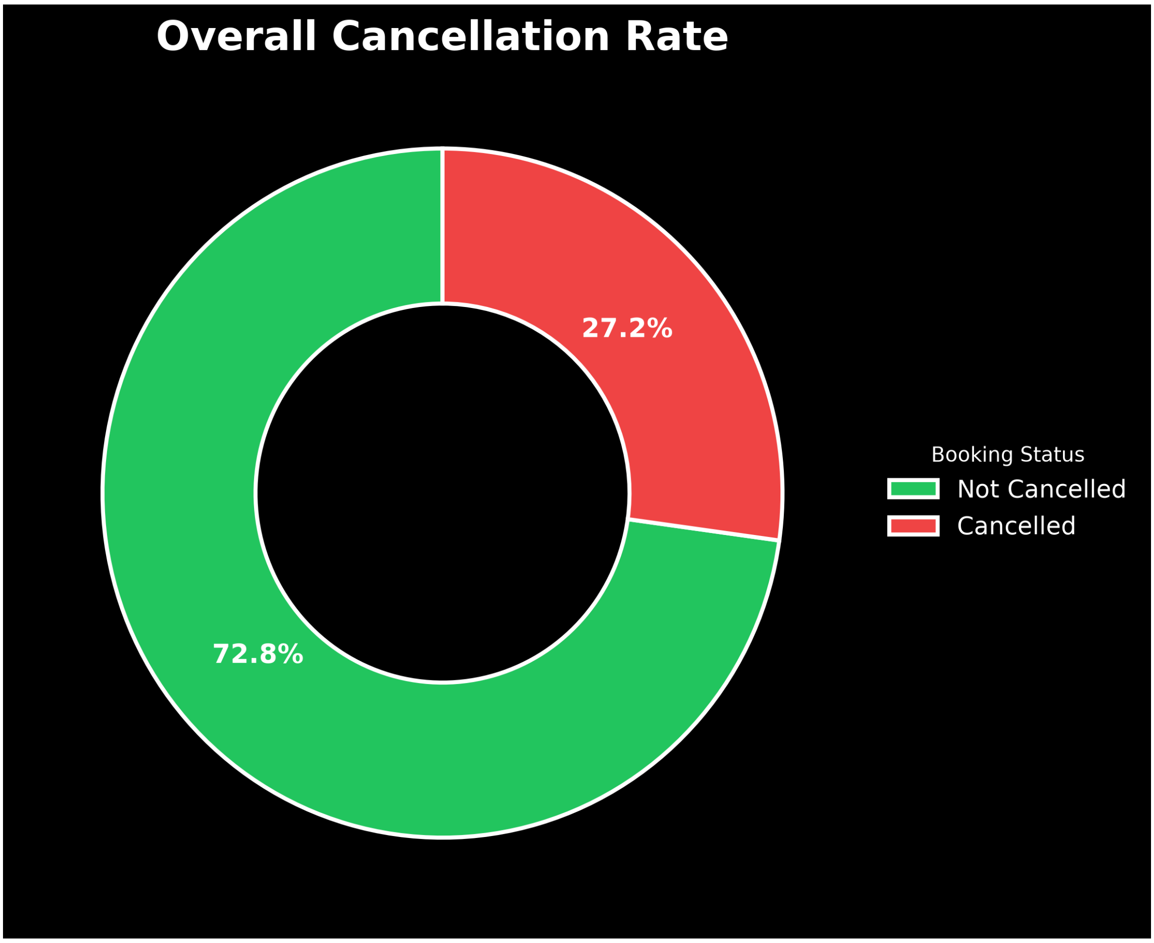


Online TA generated the highest number of bookings (49,768), followed by **Offline TA/TO** (13,801), **Direct** (11,131), and **Groups** (4,895). This indicates that online travel agencies are the primary booking channel, emphasizing their importance in hotel distribution strategies.

4.5 Cancellation Analysis

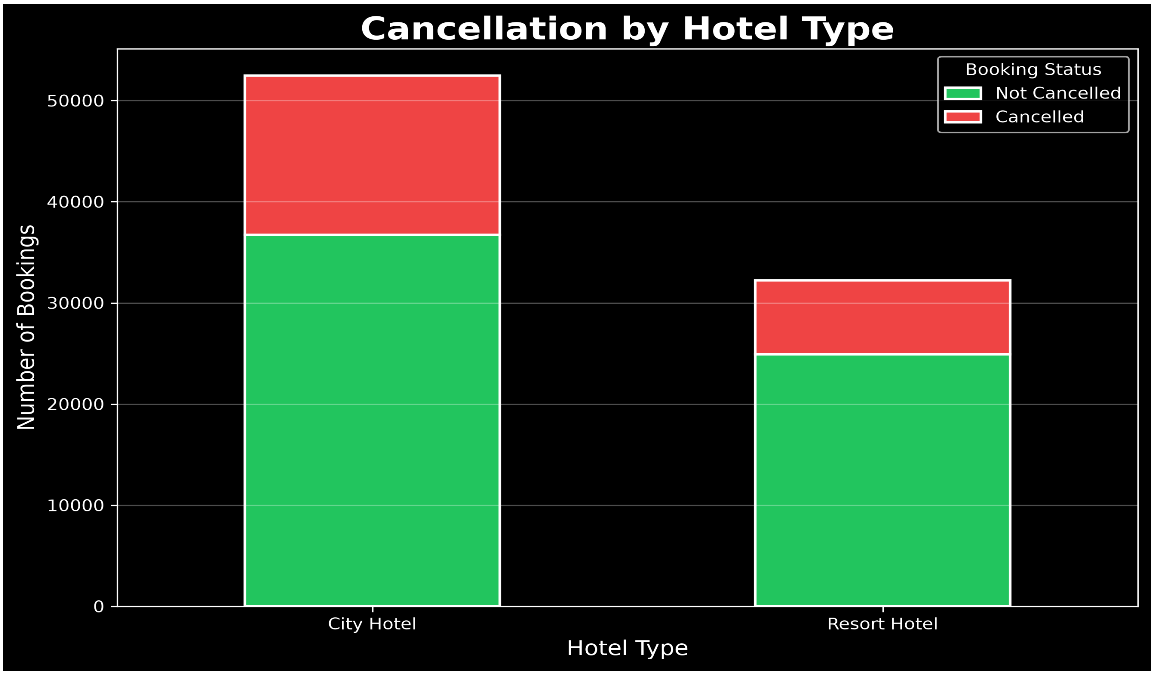
Cancellation analysis helps identify booking cancellation patterns and understand factors influencing customer booking decisions. This analysis evaluates overall cancellation rate, cancellation behavior by hotel type, market segment contribution, and the relationship between lead time and cancellations.

5.1 Overall Cancellation Rate



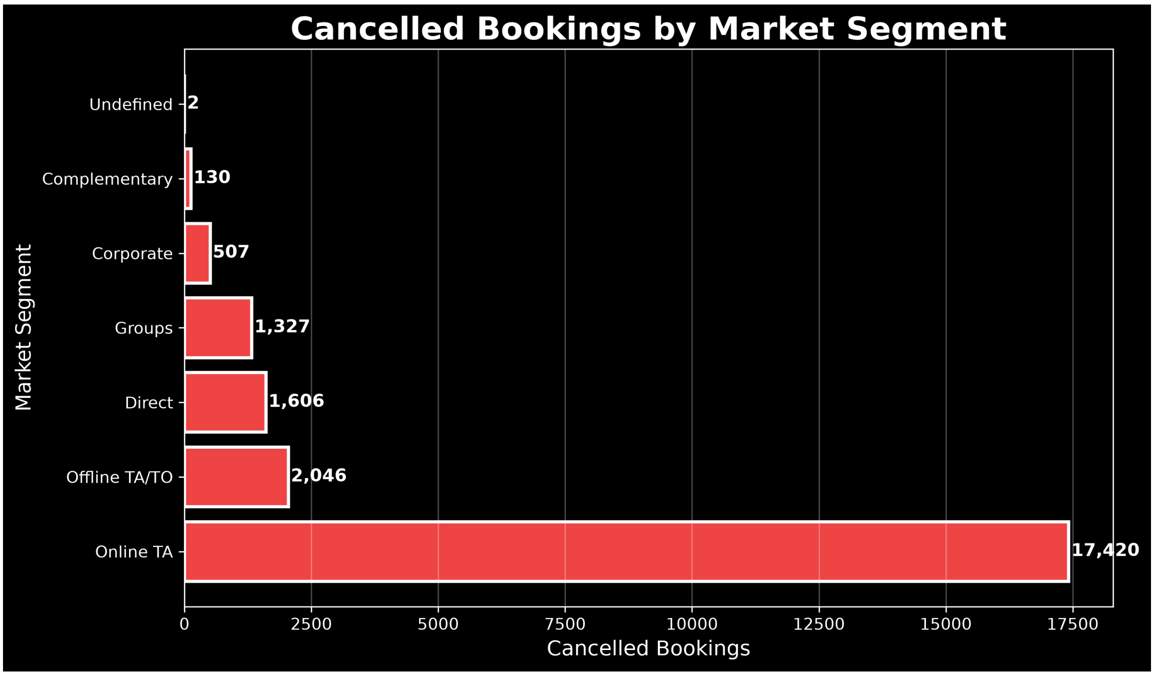
The dataset contains **61,667** successful bookings and **23,038** cancelled bookings. The cancellation rate is approximately **27.20%**, indicating that a significant proportion of reservations were cancelled before arrival.

5.2 Cancellation by Hotel Type



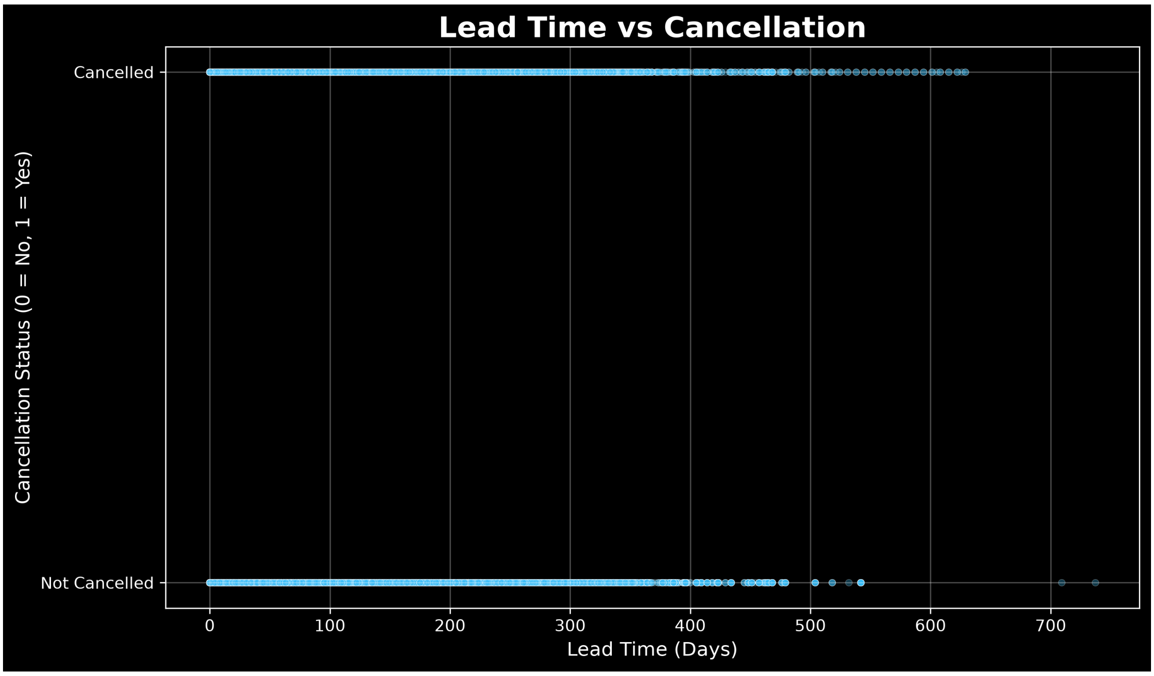
Cancellation behavior differs between hotel categories. **City Hotel** recorded the highest number of cancellations with **15,754** cancelled bookings compared to **7,284** cancellations in **Resort Hotel**. This suggests that city hotels experience greater booking uncertainty, possibly due to business travel patterns and flexible reservation behavior.

5.3 Cancellation by Market Segment



Among different booking channels, **Online TA** contributed the highest number of cancelled bookings (17,420), followed by **Offline TA/TO** (2,046) and **Direct bookings** (1,606). The high cancellation volume from Online Travel Agencies indicates that hotels should focus on improving OTA booking policies and cancellation management strategies.

5.4 Lead Time vs Cancellation Analysis

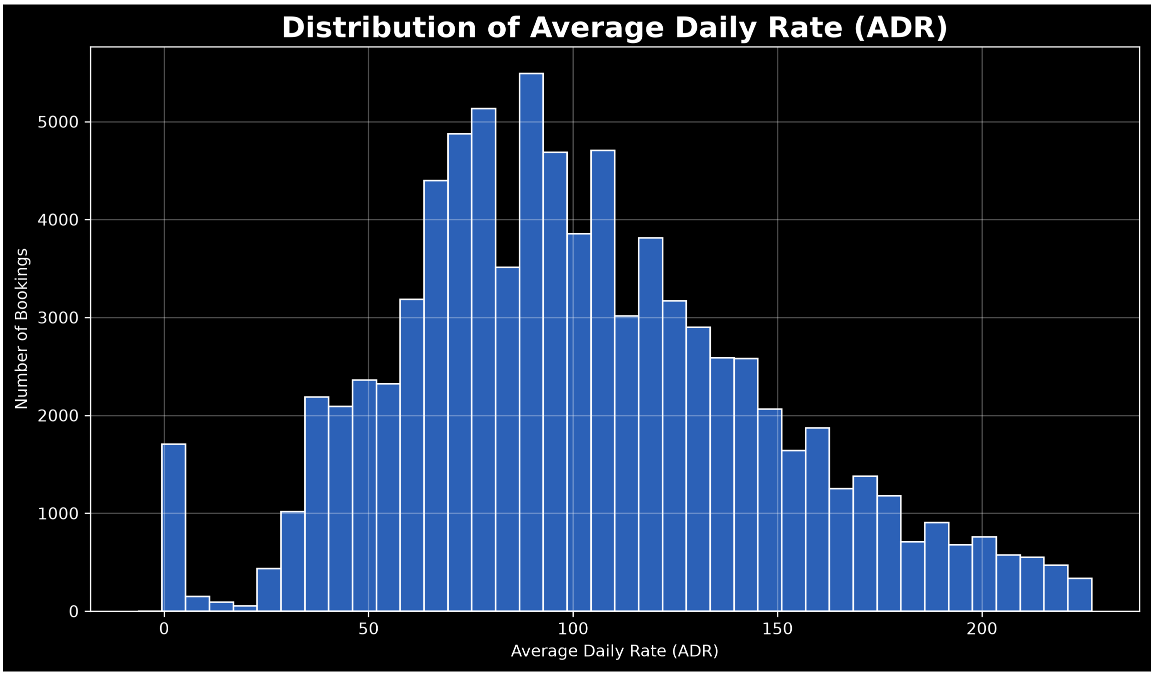


Lead time analysis shows a clear difference between cancelled and completed bookings. Cancelled reservations had an average lead time of **106.36** days, whereas non-cancelled bookings had an average lead time of **70.48** days. This indicates that bookings made far in advance have a higher probability of cancellation, highlighting the importance of effective reservation policies and demand forecasting.

6 Revenue Analysis

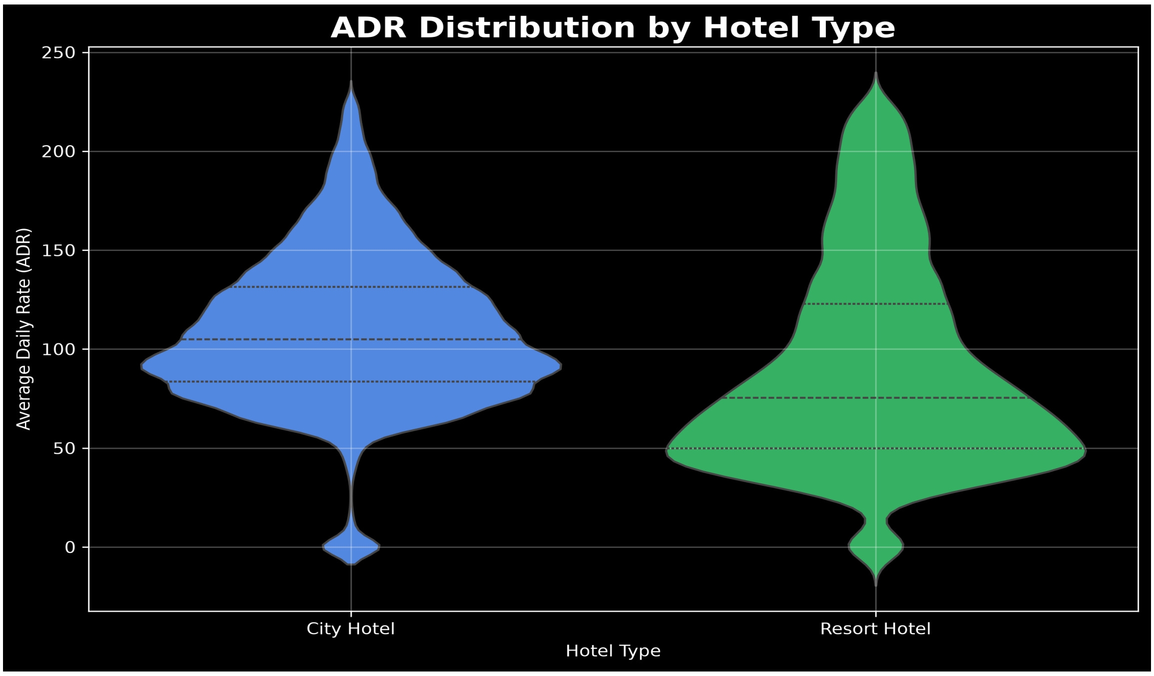
Revenue analysis focuses on understanding hotel pricing patterns and revenue generation factors. This section analyzes Average Daily Rate (ADR) distribution, revenue contribution by hotel type, seasonal pricing variations, and revenue differences between weekday and weekend stays.

6.1 Average Daily Rate (ADR) Distribution



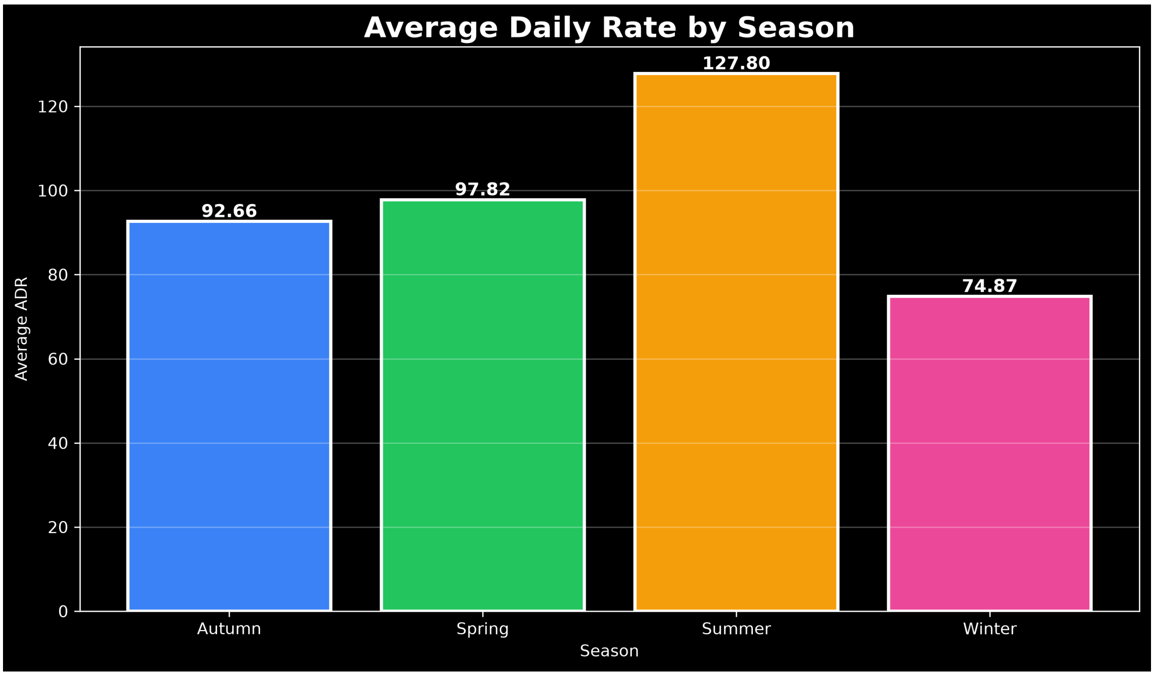
Average Daily Rate (ADR) represents the average revenue earned per occupied room per day. The analysis shows that the average ADR is **101.94**, with a median value of **96.90**. The ADR values range from **-6.38** to **226.84**, indicating variations in pricing based on hotel type, season, and booking conditions.

6.2 Revenue by Hotel Type



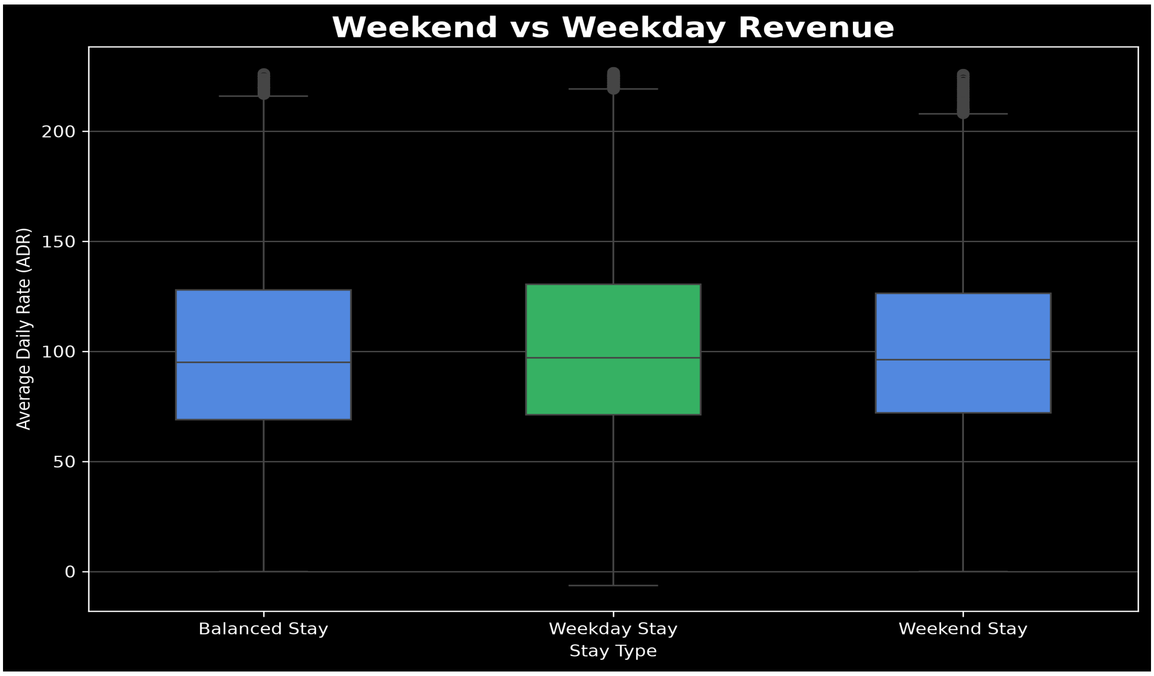
Revenue performance differs between City Hotel and Resort Hotel. **City Hotel** generated a higher average ADR of **109.05** compared to **90.35** for Resort Hotel. The higher ADR in City Hotels indicates stronger pricing power, likely due to business demand and urban location advantages. Resort Hotels show wider ADR variation because of seasonal demand patterns.

6.3 Revenue by Season



Seasonal analysis highlights significant variations in hotel pricing. **Summer** recorded the highest average ADR of **127.80**, followed by **Spring** with **97.82**. **Winter** had the lowest average ADR at **74.87**, indicating lower demand during colder months. This suggests that hotels can optimize pricing strategies according to seasonal demand fluctuations.

6.4 Weekday vs Weekend Revenue Analysis

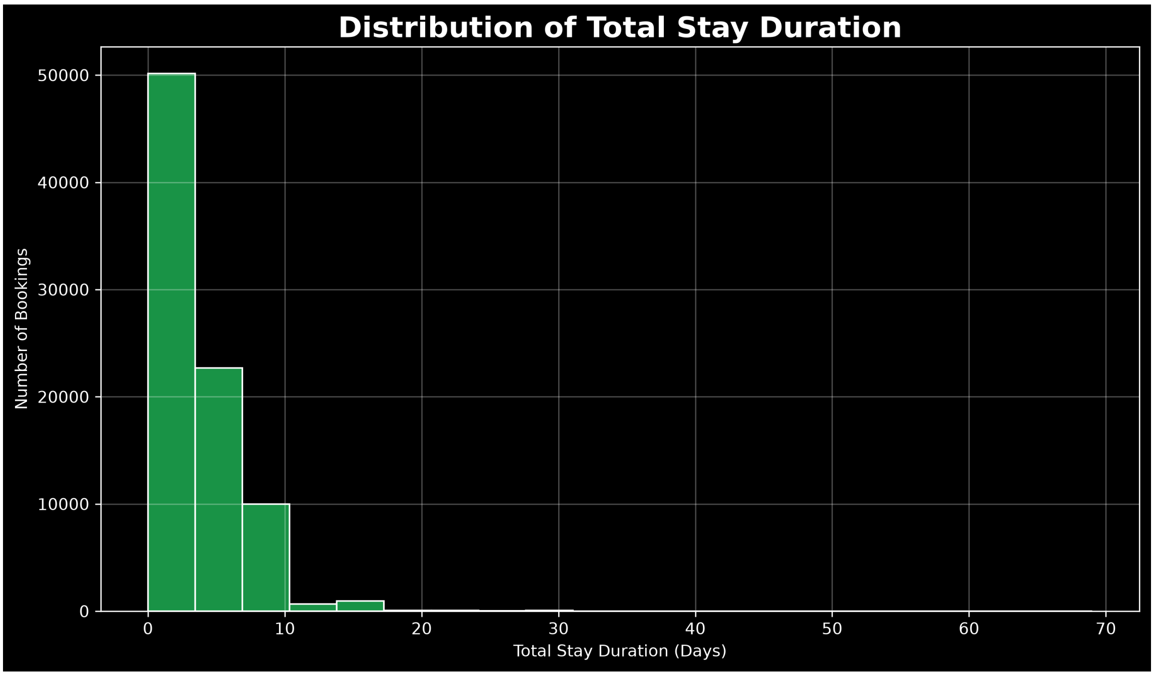


Revenue comparison between weekday and weekend stays shows slight differences in pricing. **Weekday stays** achieved the highest average ADR of **102.76**, while **Weekend stays** recorded an average ADR of **100.86**. The results indicate that weekday bookings contribute strongly to revenue generation, possibly due to business-related travel demand.

7 Stay Analysis

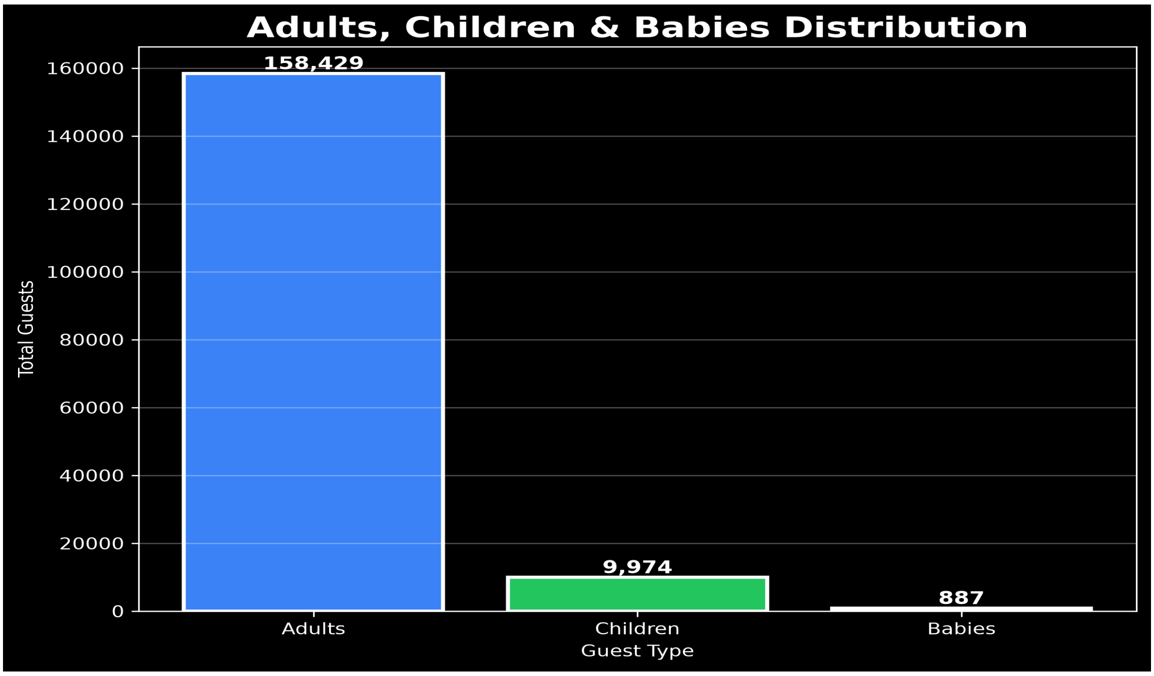
Stay analysis helps understand customer accommodation patterns and booking behavior. This analysis evaluates the distribution of stay duration, guest composition including adults, children, and babies, and the impact of special requests on hotel bookings. These insights help hotels optimize room planning, service offerings, and guest experience strategies.

7.1 Stay Duration Distribution



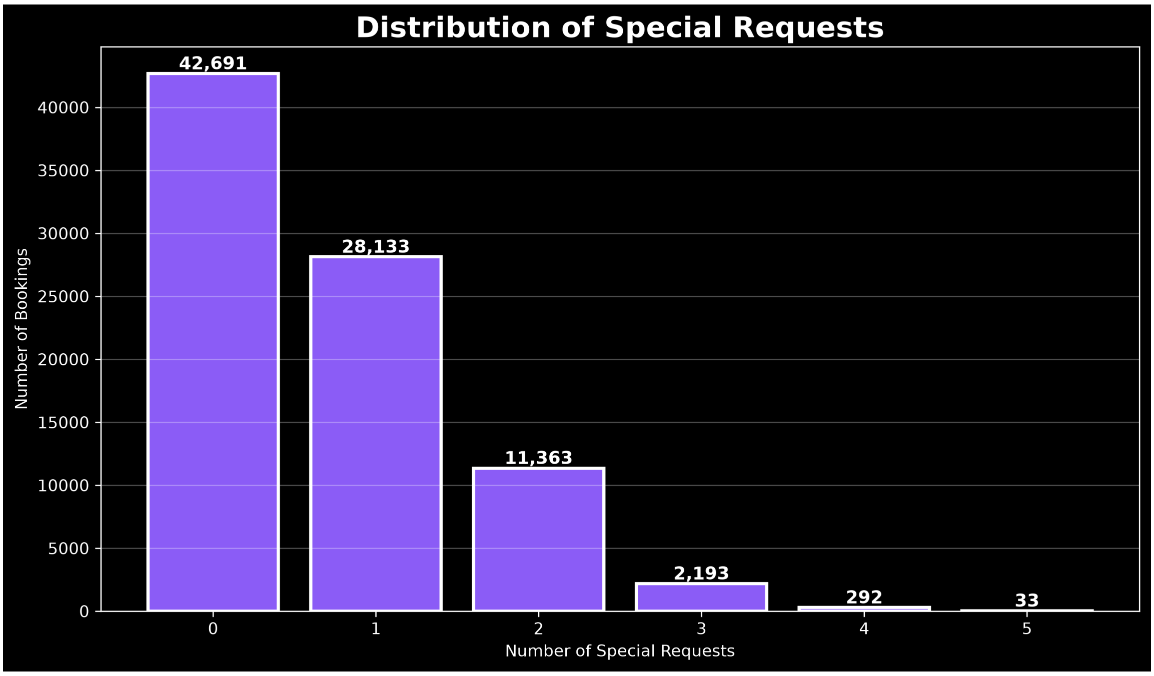
The analysis of total stay duration shows that customers stayed for an average of **3.61** days. The median stay duration was **3** days, indicating that most customers preferred short stays. The duration ranged from **0** to **69** days, showing variations in customer travel requirements.

7.2 Guest Type Distribution



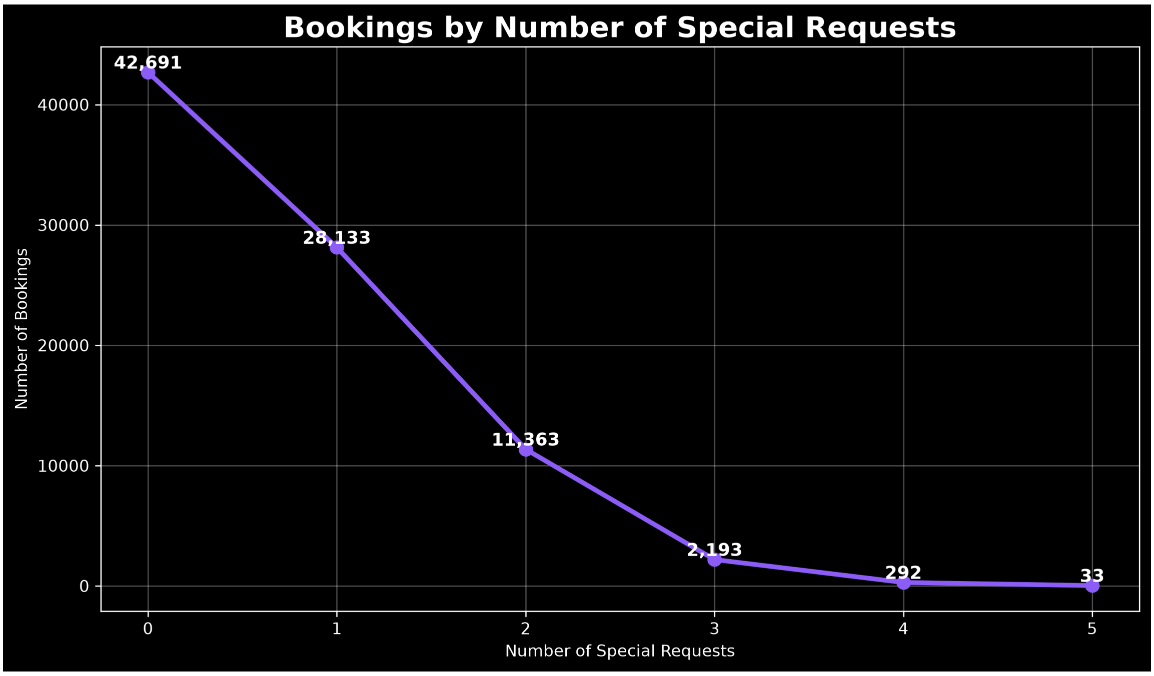
Guest composition analysis shows adults represent the majority of hotel visitors with a total count of **158,429**. Children accounted for **9,974** guests, while babies represented **887**. This indicates dataset mainly consists of adult travelers, with comparatively fewer family bookings involving children and infants.

7.3 Special Requests Analysis



Special request analysis helps identify customer preferences during hotel stays. The majority of bookings had no special requests (**42,691** bookings), while **28,133** bookings included one special request. The number of bookings decreases as the number of requests increases, indicating that most customers require only limited additional services.

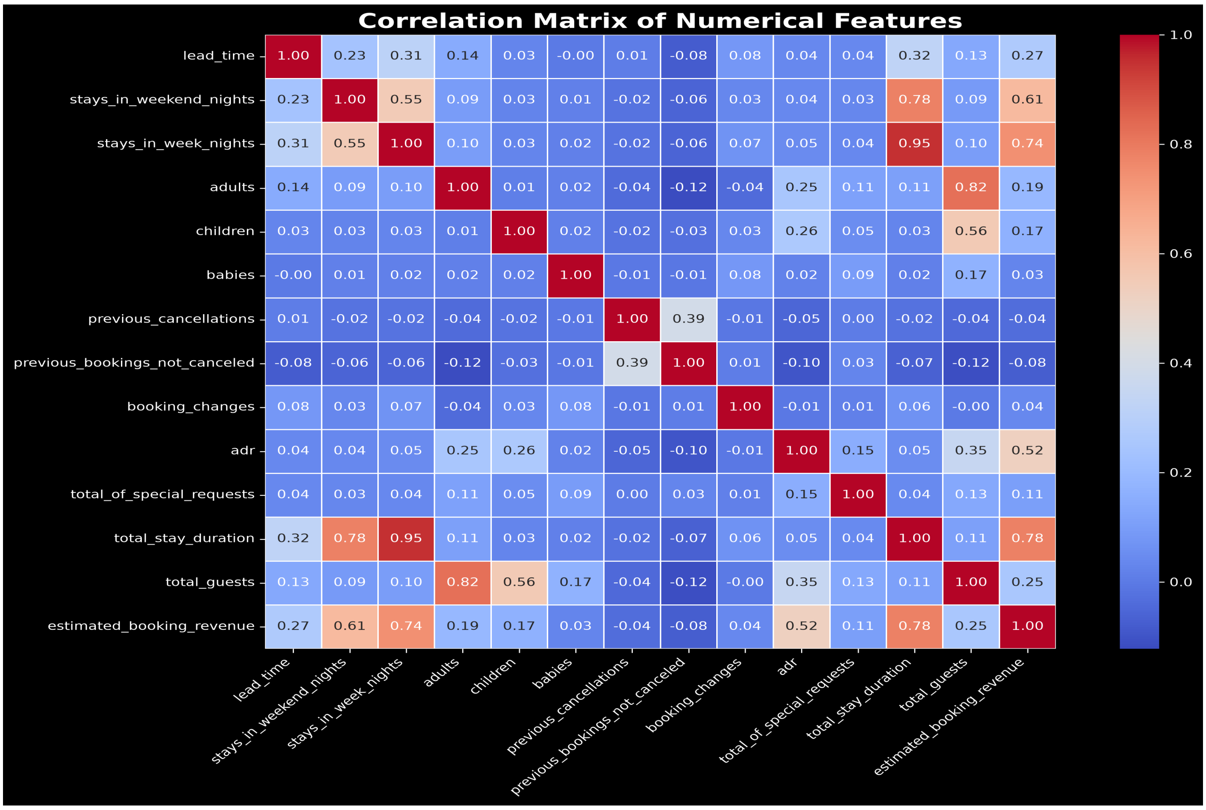
7.4 Special Requests Trend Analysis



The trend analysis shows the relationship between the number of special requests and booking frequency. Bookings are highest among customers with fewer special requests, while reservations with multiple requests are comparatively lower. This indicates that most guests prefer standard accommodation services, whereas a smaller segment requires personalized arrangements.

8 Correlation Analysis

Correlation analysis measures the relationship between numerical variables using the Pearson correlation coefficient (r), which ranges from -1 to +1. It helps identify feature relationships and supports feature selection for further statistical analysis.



The heatmap shows relationships between numerical variables. The strongest positive correlation was between **stays_in_week_nights** and **total_stay_duration** (**0.95**), followed by **total_stay_duration** and **estimated_booking_revenue** (**0.78**). Negative correlations were weak, with the highest negative relationship between **previous_bookings_not_canceled** and **total_guests** (**-0.12**).

Key Business Insights

• Booking behaviour varies based on hotel type, market segment, and seasonal demand patterns.

• Cancellation rates are influenced by factors such as lead time, hotel type, and booking channels.

• Revenue performance depends on ADR variations across hotels, seasons, and stay patterns.

• Longer stays and higher guest counts contribute significantly to increased booking revenue.

• Online Travel Agencies generate high booking volumes but also contribute to a large share of cancellations.

• Seasonal pricing strategies can help hotels maximize revenue during high-demand periods.

• Customer preferences such as stay duration and special requests provide opportunities for personalized services.

• Statistical analysis confirms meaningful relationships among booking, customer, and revenue-related factors.